

CARDIOVASCULAR DISEASE (CVD) RISK: DIAGNOSIS

CVD risk is the chance of having a heart attack or stroke over the next 10 years

Step 1

Identify if the patient has established CVD:

- If patient has had previous heart attack, stroke or TIA or is known with angina (ischaemic heart disease) or peripheral vascular disease, manage as CVD → 129.
- If current/recent chest pain, especially on exertion and relieved by rest, consider **ischaemic heart disease** → 137.
- If current/recent leg pain, especially on walking and relieved by rest, consider **peripheral vascular disease** → 139.
- If new sudden asymmetric weakness or numbness of face, arm or leg; difficulty speaking or visual disturbance: consider **stroke** or **TIA** → 136.

Step 2

Look for modifiable CVD risk factors:

- Ask about **smoking**: consider the patient who quit smoking in the past year a smoker for CVD risk assessment.
- Calculate **BMI**: weight (kg) ÷ height (m) ÷ height (m). A BMI > 25 is a risk factor.
- Measure **waist circumference** while standing or breathing out, midway between lowest rib and top of iliac crest. More than 80cm (woman) or 94cm (man) is a risk factor.
- Look for **hypertension**: check BP. If BP ≥ 140/90 and not known with hypertension → 132.
- Look for **diabetes**: if not known with diabetes, check glucose → 17.

Step 3

Calculate the patient's CVD risk if no established CVD:

- If total and HDL cholesterol results available, calculate the patient's 10-year CVD risk score below or use the EMGuidance app tool.
- If cholesterol results are not available, use instead the chart that calculates CVD risk using BMI → 128.

If no established CVD and cholesterol available, calculate the patient's 10-year CVD risk using the scoring system below

- Calculate CVD risk score by adding the points in each of the tables below, using patient's age, sex, total cholesterol, HDL cholesterol, BP, smoking status and diabetes status:
 - If **man**: if score < 11, then CVD risk is < 10%. If score 11-14, then CVD risk is 10-20%. If score ≥ 15, then CVD risk is > 20%.
 - If **woman**: if score < 13, then CVD risk is < 10%. If score 13-17, then CVD risk is 10-20%. If score ≥ 18, then CVD risk is > 20%.

Age (years)	Man	Woman
35-39	2	2
40-44	5	4
45-49	6	5
50-54	8	7
55-59	10	8
60-64	11	9
65-69	12	10
70-74	14	11
75-79	15	12

Total cholesterol (mmol/L)	Man	Woman
< 4.1	0	0
4.1-5.19	1	1
5.2-6.19	2	3
6.2-7.2	3	4
> 7.2	4	5

HDL cholesterol (mmol/L)	Man	Woman
> 1.5	-2	-2
1.3-1.49	-1	-1
1.2-1.29	0	0
0.9-1.19	1	1
< 0.9	2	2

Systolic BP (mmHg)	Man		Woman	
	Not on BP treatment	On BP treatment	Not on BP treatment	On BP treatment
< 120	-2	0	-3	-1
120-129	0	2	0	2
130-139	1	3	1	3
140-149	2	4	2	5
150-159	2	4	4	6
≥ 160	3	5	5	7

	Man	Woman
Smoker	4	3
Diabetes	3	4

Step 4

Explain to the patient what his/her risk of heart attack or stroke might be over next 10 years:

- If CVD risk is < 10%, there is a less than 1 in 10 chance that s/he may have a heart attack or stroke over the next 10 years.
- If CVD risk is 10-20%, there is a 1-2 in 10 chance that s/he may have a heart attack or stroke over the next 10 years.
- If CVD risk is > 20%, there is a more than 2 in 10 chance that s/he may have a heart attack or stroke over the next 10 years.

Step 5

Use the patient's CVD risk to decide treatment and frequency of follow-up:

- If CVD risk factor or a CVD risk ≥ 10%, manage the CVD risk → 129.
- If CVD risk < 10% and no CVD risk factors, reassess CVD risk after 5 years.